

Markov Chains: Stationary Distributions

1. **Stationary distribution:** Let X_n be a Markov Chain having state-space S and transition function P . If $\pi(x)$ is a probability distribution such that:

$$\sum_{x \in S} \pi(x) P(x, y) = \pi(y), \quad y \in S,$$

then π is a stationary distribution. This condition is often referred to as *general balance* and written down as $\pi^T P = \pi^T$.

2. **What the above means:** P maintains π . That is, if $\pi_0 = \pi$, then $\pi_t = \pi_0 P^t = \pi$ for all t .
3. **Our goal:** Under certain conditions (irreducible and aperiodic - we will see these later), π is unique and $\pi_t \rightarrow \pi$ as $t \rightarrow \infty$, irrespective of π_0 .
4. **Condition on transition probabilities:** Suppose that a stationary distribution, π , exists and that:

$$\lim_{n \rightarrow \infty} P^n(x, y) = \pi(y), \quad y \in S.$$

So,

$$P(X_n = y) = \sum_{x \in S} \pi_0(x) P^n(x, y).$$

Therefore,

$$\lim_{n \rightarrow \infty} P(X_n = y) = \sum_{x \in S} \pi_0(x) \pi(y) = \pi(y).$$

Thus, $X_n \xrightarrow{L} \pi$.

5. **Observe:** The stationary distribution, when $X_n \xrightarrow{L} \pi$, is unique.
6. **Example:** For the 2-state Markov Chain we saw that, if $p + q > 0$, then the stationary distribution is given by:

$$\pi(0) = \frac{q}{p+q}; \pi(1) = \frac{p}{p+q}.$$

We also saw that $0 < p + q < 2$ then $\lim P^n = \pi$ holds.

7. **Example:** Consider a Markov Chain having state-space $S = \{0, 1, 2\}$ and the transition matrix:

$$P = \begin{bmatrix} 1/3 & 1/3 & 1/3 \\ 1/4 & 1/2 & 1/4 \\ 1/6 & 1/3 & 1/2 \end{bmatrix}.$$

To check whether this chain has a stationary distribution or not, we try solving the general balance equations and check if the solution is a probability distribution. So, for the above:

$$\pi^T (I - P) = 0$$

which gives:

$$\pi^T = \left(\frac{6}{25}, \frac{2}{5}, \frac{9}{25} \right).$$

This is the unique stationary distribution.

8. **HW:** Consider an irreducible birth and death chain on $\{0, 1, \dots, d\}$ or on the non-negative integers (here $d = \infty$). Investigate the finite and infinite cases for the existence of the stationary distribution. When it exists, give the form of the stationary distribution.

9. **Example:** Consider the Ehrenfest chain with $d = 3$ balls. Then,

$$P = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1/3 & 0 & 2/3 & 0 \\ 0 & 2/3 & 0 & 1/3 \\ 0 & 0 & 1 & 0 \end{bmatrix}.$$

You may identify this as an irreducible birth and death chain and use your results from the above HW to get its stationary distribution. Otherwise, you may directly solve the general balance equation to get:

$$\pi^T = \left(\frac{1}{8}, \frac{3}{8}, \frac{3}{8}, \frac{1}{8} \right).$$

Note, however, that $\lim P^n \nrightarrow \pi$, since $P^n(x, x) = 0$ for all odd n .

10. **Modified Ehrenfest Chain:** Suppose we have 2 boxes labelled 1 and 2 and d balls labelled $1, 2, \dots, d$. Initially some of the balls are in box 1 and the remainder are in box 2. An integer is selected at random and the ball labelled by that integer is removed from its box. We now

select at random one of the two boxes and put the removed ball into this box. The procedure is repeated indefinitely, the selections being made independently. Let X_n be the number of balls in box 1 after the n^{th} trial. Set $d = 3$. Then, the transition matrix is given by:

$$P = \begin{bmatrix} 1/2 & 1/2 & 0 & 0 \\ 1/6 & 1/2 & 1/3 & 0 \\ 0 & 1/3 & 1/2 & 1/6 \\ 0 & 0 & 1/2 & 1/2 \end{bmatrix}.$$

The stationary distribution is again:

$$\pi^T = \left(\frac{1}{8}, \frac{3}{8}, \frac{3}{8}, \frac{1}{8} \right).$$

We will see later that here $\lim P^n \rightarrow \pi$.

11. **Laying out the path in front:** Consider an irreducible birth and death chain with stationary distribution π . Suppose that $P(x, x) = r_x = 0$, $x \in S$, as in the Ehrenfest chain. Then, at each transition the chain moves either one step to the right or one step to the left. So, the chain can return to its starting point only after an even number of transitions. Thus, $P^n(x, x) = 0$ for all n , and for such a chain the formula

$$\lim_{n \rightarrow \infty} P^n(x, y) = \pi(y)$$

clearly fails to hold. But, there is a way to handle such situations. Recall our old friend Mr. Cesaro from Real analysis: Let $\{a_n\}$ be a sequence of numbers such that, $\lim a_n = L$ for some finite number L , then

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n a_i = L.$$

However, this formula can still hold even when $\lim a_n$ fails to exist. For example, if $a_n = 0$ for odd n and $a_n = 1$ for even n , then a_n has no limit, but the means still converge with $L = 1/2$. We will see see that

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n P^k(x, y)$$

will always exist for an arbitrary Markov Chain. Then we will use this limit to determine which Markov Chains have stationary distributions and when there is such a stationary distribution.